Md Abu Sayeb Rayhan

Computer Vision · Reliable AI · Robot Perception

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Google Scholar · ResearchGate · GitHub · LinkedIn

Seeking a funded MS or PhD, 2026–27 · International applicant · IELTS 7.0 (CEFR C1)

Research Profile

Computer-vision researcher with **seven papers (five as first author)** in reliable medical AI, robot perception and image forensics, and **18 months building production video analytics in Japan**. My work asks whether a model's output can be trusted where it is deployed, from uncertainty fusion that flags inputs a classifier should not answer to preregistered closed-loop tests of robot perception.

Research Interests

- **Reliable & uncertainty-aware vision**: out-of-distribution detection, evidential and conformal methods
- **Video understanding**: multi-object tracking, activity recognition, pose estimation
- **Robot perception**: closed-loop evaluation of perception for navigation
- **Generative models**: diffusion models and synthetic-image forensics
- **Efficient perception**: vision under the latency and power budgets of edge hardware

Education

B.Sc. in Computer Science & Engineering, Comilla University, Bangladesh — Jan 2019 – May 2024
GPA 3.40 / 4.00
Thesis: Text-to-Image Generation using Diffusion Models – fine-tuned Stable Diffusion on a purpose-built dataset of Bangladeshi landmarks; supervised by Partha Chakraborty.
Relevant coursework: Artificial Intelligence, Algorithm Design, Theory of Computation, Introduction to Data Science.

Publications

Seven papers, five as first author (my name in bold). Abstracts, figures and slides: abusayeb.github.io/research

Peer-reviewed conference papers

[1] **M. A. S. Rayhan**, M. Fahimah, M. T. Mahmud, H. Hiya, A. S. Dhar. "RGEF: Reliability-Gated Evidential Fusion for Robust Medical AI." *IEEE Int. Conf. on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, 2026. *In press.* [paper page]

[2] **M. A. S. Rayhan**, A. Shutradhar, H. Hiya. "Forensic Invariant Learning for Synthetic Image Detection: Bridging Benford's Law and Topological Analysis with Machine Learning Ensembles." *IEEE Int. Conf. on Quantum Photonics, Artificial Intelligence and Networking (QPAIN)*, 2026. [IEEE Xplore]

[3] A. Shutradhar, **M. A. S. Rayhan**, M. Khatun, B. Das. "BD Eye-Doc: When Computer Vision Meets Conversational AI for Scalable Eye Disease Screening in Bangladesh." *IEEE QPAIN*, 2026. [IEEE Xplore]

[4] A. S. Dhar, B. Das, M. M. Hassan, **M. A. S. Rayhan**, H. T. Saydia, S. A. Shimu. "Is CNN Dead? A Comprehensive Empirical Analysis of Convolutional Neural Networks, Ensemble Methods, and Vision Transformers for Rice Leaf Disease Classification in Low-Resource Agricultural Settings." *Int. Conf. on Emerging Computing and Communication Technologies (ECCT)*, 2026; Advances in Computer Science Research, Atlantis Press (Springer Nature). *In press.*

Journal manuscripts under review

[5] **M. A. S. Rayhan**, H. Hiya, A. S. Dhar. "Does learning when to look pay for home service robots? A preregistered test of adaptive perception scheduling in closed-loop navigation." Submitted to *Intelligent Service Robotics* (Springer), 2026.

[6] **M. A. S. Rayhan**, H. Hiya, A. S. Dhar. "Doorways set the pace of robot vacuum cleaning: a parameter-free room-graph model, tested and confirmed in 200 real homes." Submitted to *Robotics and Autonomous Systems* (Elsevier), 2026.

[7] **M. A. S. Rayhan**, H. Hiya, A. S. Dhar. "Input-conditional uncertainty fusion for skin lesion screening." Submitted to *Biomedical Signal Processing and Control* (Elsevier), 2026.

Research Experience

Independent Researcher – Dhaka, Bangladesh — Apr 2026 – present
With H. Hiya (Jahangirnagar University) and A. S. Dhar (Comilla University)

- Designed and ran a **preregistered closed-loop study** of adaptive perception for home robots [5] in 25 held-out scanned homes (Habitat, HM3D) against 12 baselines; the learned scheduler cut detector compute by 85% but missed its preregistered success and safety bar, and the paper reports that negative result.
- Derived and tested a **parameter-free room-graph model** of robot-vacuum cleaning [6] over 17,580 simulated runs in 200 homes; a second preregistration on 100 unseen homes confirmed it ($R^2 = 0.81$).

- Extended RGEF into a journal study [7]: per-image reliability gating of eight uncertainty signals on a deduplicated external archive raised AUROC from 0.6945 (static fusion) to 0.7946.

Research Assistant – Dept. of CSE, Comilla University (supervisor: P. Chakraborty) Jan 2023 – Mar 2024

- Designed datasets, ran experiments and analysed results for empirical studies in human–robot interaction and optical character recognition; results were carried into peer-reviewed conference papers.

Professional Experience

Data Scientist – Othello Financial Services Co. Ltd., Osaka, Japan Oct 2024 – Mar 2026
AGiFT AI Camera team

- Built multi-object tracking (ByteTrack), pose estimation (MoveNet) and activity recognition (MMAction2) pipelines that analysed shopper behaviour on live retail video.
- Clustered pose output with OPTICS to model group dynamics, and parallelised inference to cut processing latency by **about 70%**.
- Fine-tuned a YOLO detector (90%+ accuracy) for a video event-segmentation system that automated sports highlight extraction and cut production cost by **50%**; generalised it to user video from festivals and concerts.
- Deployed a retrieval-augmented generation system (LangChain, Pinecone, GPT-4) on Microsoft Azure, and connected in-store IoT edge devices to the real-time vision pipelines.

AI Engineer (contract) – Government of Bangladesh project, Cumilla Feb 2024 – Jul 2024

- Led a three-person team that delivered a bilingual (Bangla/English) document-search system combining OCR with dense embeddings across PDF, Word, Excel and image files, shipped as an offline desktop application (PyQt, ChromaDB).

Founder and developer – eGP AutoPilot, Dhaka (alongside independent research) 2026 – present

- Designed, built and released a bilingual desktop application that automates tender submission on Bangladesh's national e-Government Procurement portal (Playwright, PyQt, Firebase); v2.2.0 ships on macOS and Windows with 273 offline tests.

Leadership & Service

- **Team lead**, Government of Bangladesh project (2024): led a team of three from design to delivery.
- **Joint General Secretary**, North Bengal Student Association, Comilla University (2023–2024): elected; organised events and student networking and took part in policy discussions.
- **Batch Representative**, Dept. of CSE, Comilla University (2022–2024): elected by peers to represent the cohort to faculty and administration on academic quality and student welfare.
- Took part in student-led initiatives promoting technology awareness and collaborative learning at Comilla University (2019–2024).

Honours, Awards & Presentations

- **Best Presenter Award**, IEEE QPAIN 2026.
- Conference presentations: IEEE RAAICON 2026 (RGEF, slides); IEEE QPAIN 2026 (two papers); ECCT 2026.
- Inter-University Programming Contest, Comilla University (2019–2021); certified in Mobile Game & Application Development, ICT Division, Government of Bangladesh (2022).
- Inter-department sports, Comilla University (2019–2023): cricket champions; third place in volleyball.

Skills & Languages

- **Methods:** uncertainty quantification, out-of-distribution detection, evidential deep learning, conformal prediction, object detection, multi-object tracking, action recognition, pose estimation, diffusion models, retrieval-augmented generation, OCR.
- **Research practice:** preregistration, closed-loop simulation (Habitat-sim, HM3D), Friedman and Holm-corrected tests, reproducible result pipelines.
- **Tools:** Python (advanced), C/C++, SQL; PyTorch, TensorFlow, OpenCV, MMAction2, YOLO, Hugging Face, scikit-learn, LangChain; Azure, Docker, Git, LaTeX.
- **Languages:** Bangla (native); English (C1; IELTS 7.0 – Listening 7.5, Reading 8.0, Writing 6.5, Speaking 6.5); Hindi (spoken, fluent); Japanese (spoken B1, written A2; studying toward JLPT N3).

References

Contact details available on request.

Partha Chakraborty
Associate Professor, CSE, Comilla University (thesis and research supervisor)

Md Mohibullah
Assistant Professor, CSE, Comilla University; PhD Fellow, Saitama University

Hossain Md Erfan
Project Manager, Othello Financial Services, Japan (direct supervisor)